Puneet Velidi

Statistical AI/ML Researcher · Founder, ChimeraML

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Education

University of Victoria — **Victoria, BC**
M.Sc. in Statistics (thesis-based, fully funded) — *Jan. 2025 – Dec. 2026*
Co-supervisors: Prof. Farouk Nathoo & Prof. Michelle Miranda. Thesis: *Methods for Censored Covariates, Spatial Transcriptomic Data and Neural Manifolds*.

Cornell University — **Ithaca, NY**
B.A. in Statistics & Economics — *Sep. 2018 – Dec. 2022*

Publications († joint first authorship)

P. Velidi, M. Miranda, F. Nathoo, A. Mbilinyi, and C. Beaulac. *Shape Analysis as an Inductive Bias for Glioma Grading from Tumor Contours*. *Submitted for publication*. **2026**. arXiv

P. Velidi, E. Amico, F. Nathoo, and M. Miranda. *Neural manifold connectomics reveals multiregime functional connectivity*. **Under review**. **2026**. bioRxiv

P. Velidi, M. Griffin, J. Liu, et al., F. Nathoo. *SPARTACCUS: Semiparametric Bayesian Proportional Hazards Model with a Censored Covariate in Huntington's Disease*. *In preparation*. **2026**.

P. Velidi, Z. Wei, and F. Nathoo. *Covariance Nonstationarity is Evident in Spatial Transcriptomics and Provides a New Categorization of Spatially Varying Genes*. **Under review**. **2026**. bioRxiv

F. Luchese, **P. Velidi**, et al., P. Cassano. *Transcranial Photobiomodulation on Language and Cognitive Performance in Down Syndrome: A Pilot Randomized Sham-Controlled Trial*. **Under review**. **2026**. medRxiv

B. Ferreras ... **P. Velidi**, et al., P. Cassano. *Transcranial Photobiomodulation in Bipolar Disorder: A Case Series on Executive Skills and fMRI BOLD Signal*. *Submitted for publication*. **2026**. SSRN

G. Gonzalez-Garibay, **P. Velidi**, et al., P. Cassano. *Volumetric Changes Induced by Transcranial Photobiomodulation Therapy in Major Depressive Disorder: An Exploratory Neuroimaging Study*. *Submitted for publication*, 2026. SSRN preprint

P. Velidi†, Z. Wei†, et al., F. S. Nathoo. *Generalized Jeffreys's Approximate Objective Bayes Factor: Model Selection Consistency, Finite-Sample Accuracy, and the Bayesian Evidence in 71,126 Clinical Trial Findings*. **Under revision**. **2025**. arXiv.

S. P. Singleton†, **P. Velidi**†, et al., A. Kuceyeski. *Altered structural connectivity and functional brain dynamics in individuals with heavy alcohol use elucidated via network control theory*. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*, **2024**.

Academic Awards & Scholarships

Labs4 Technology Readiness Level-Up Fellowship — 15,000 CAD (2025).
UVic Betty & Gilbert Kennedy Graduate Scholarship in Mathematics & Statistics — 5,130 CAD (2025-2026).
Maud Menten Institute / PIMS Research Accelerator Award — 5,000 CAD (2025).
UVic FGS Graduate Fellowship, Master's — 5,000 CAD (2025-2026).

Selected Industry Experience

Founder — ChimeraML — **Vancouver, BC & Washington, DC**
Deeptech research studio — *Sept. 2025 – Present*
- Backed by NVIDIA Inception, Labs4, and Mitacs.

Software Engineer II — Walmart Global Tech — **Bentonville, AR**
Enterprise Business Services – Associate Digital Experience — *Feb. 2023 – May 2024*
- Early Career Recognition, awarded to <10% of the 2023 Walmart Global Tech Development Program.

Research Experience

Graduate Researcher — University of Victoria, Dept. of Mathematics & Statistics — **Victoria, BC**
Nathoo Lab, PIs: Prof. Farouk Nathoo & Prof. Michelle Miranda — *Jan. 2025 – Present*

Visiting Researcher — University of Birmingham, School of Mathematics — **Birmingham, UK**
AMICO Lab, PI: Prof. Enrico Amico — *Aug. 2025*

Postbacc Researcher — Harvard Medical School & MGH, Neuropsychiatry & Neuromodulation — **Boston, MA**
MGH Photobiomodulation Laboratory, PI: Prof. Paolo Cassano — *May 2024 – Dec. 2024*
- Statistical analysis of the TransPhoM-DS, TRIADE, TPEB, and TRAP-AD clinical trials.

Undergraduate Researcher — Weill Cornell Medicine, Dept. of Radiology — **Ithaca, NY**
Computational Connectomics Laboratory, PI: Prof. Amy Kuceyeski — *Jan. 2022 – Dec. 2022*

Selected Invited Talks & Sessions

Invited Session Chair & speaker, 7th ICSA-Canada Chapter Symposium (2026) — Neural manifold connectomics reveals multiregime functional connectivity.

Invited Seminars: "Applications of Protein Folding Models in Immunotherapeutic Research," UBC Math Bio (2026); "70,000 Leagues Under the Sea of p-Values," UBC Statistics & Maud Menten Institute (2025); "Network Control Theory in Neuroscience," UVic Cognition & Brain Sciences and Maud Menten Institute (2025).

Technical Report

R. Adenane, J. Beaudin, L. Kankanamge, and **P. Velidi**. *Immunodynamic Modeling in Melanoma and Gastric Cancer*. **2025**.

Poster Presentations (full titles and posters)

Photobiomodulation and glymphatic CSF dynamics in MCI / early Alzheimer disease — Clinical Research Day, 2026 (upcoming).
Bayesian survival analysis with a right-censored covariate — ICSA-Canada, 2026.
Bayesian analysis of the TransPhoM-DS clinical trial — OHBM, 2026.
Ahab's Harpoon: calibrated irreplicability detection with approximate objective Bayes factors — SSC, 2026.
Reassessing clinical-trial evidence with extended approximate objective Bayes factors — MMI HQP Summit, 2025.
Photobiomodulation-induced volumetric brain changes in depression — CNS, 2025.
Structural connectivity and brain dynamics in heavy alcohol use via network control theory — OHBM, 2024.
Network control analysis of individuals with substance use disorder — OHBM, 2023.